MIL-C-82493(OS)
26 July 1967

Superseding
NOTS XS 557 Rev. A
12 May 1967

MILITARY SPECIFICATION

CABLE ASSEMBLY

MARK 10 MOD 0

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy

1. SCOPE

1.1 This specification establishes the requirements of one type of Cable Assembly Mark 10 Mod 0. (See 6.1 for intended use and description.)

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein except as noted.

SPECIFICATION

FEDERAL

QQ-A-225/6	Aluminum Alloy Bar, Rod, and Wire, Rolled, Drawn, or Cold Finished, 2024-T4

MILITARY

MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-A-8625	Anodic Coatings for Aluminum and Aluminum Alloys

FSC -1375

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STANDARDS

FEDERAL

FED-STD-102	Preservation, Packaging, and Packing Levels

MILITARY

MIL-STD-105D	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

DRAWINGS

BUREAU OF NAVAL WEAPONS

LD 497942	Connector, Receptacle Electrical
LD 497959	Cable Assembly Mk 10 Mod 0
LD 540588	Preserved (Packaging) Condition (Cable Assembly Mk 10 Mod 0)
LD 540589	Shipping (Packing) Condition (Cable Assembly Mk 10 Mod 0)
LD 615556	Test Set, Circuit Analyzer Cable Assembly Mk 10 Mod 0
LD 620138	Cable Test Assembly (for ASROC Cable Mk 10 Mod 0)
Dwg. 1660323	Weldment, Extractor
Dwg. 2169895	Plug, Housing Assembly

PUBLICATIONS

BUREAU OF NAVAL WEAPONS

OD 24568	Continuity and Insulation Resistance Testing of Cable Assembly Mark 10 Mod 0

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(Copies of specifications, standards, drawings and publications requested by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General requirements. Unless otherwise specified in the contract or order, all materials and treatments of materials and parts used in the manufacture of the cable shall be in accordance with the requirements specified herein and with the applicable documents listed in BuWeps LD 497959.

3.2 Preproduction samples. Unless otherwise specified in the contract or order, preproduction samples of the cable shall be manufactured using the methods and procedures proposed for production. The sample shall be tested as specified in Section 4 herein and is for the purpose of determining that, prior to starting production, the contractor' s production methods are capable of yielding items that comply with the technical requirements of the contract.

3.3 Performance and product characteristics. The cable shall meet the following performance and product characteristics.

3.3.1 Electrical continuity and circuit resistance.

3.3.1.1 Electrical continuity. The circuits of the cable shall be continuous as determined in 4.6.1.1.1.

3.3.1.2 Resistance of initiation circuit. The resistance of the initiation circuits listed below shall not exceed 0.15 ohm:

Connectors	J5		P2	Connectors	J3		P2
Pins	H	to	AA	Pins	A	to	BA
Pins	J	to	AC	Pins	B	to	BB
Pins	R	to	AB	Pins	C	to	BC

3.3.2 Insulation resistance. Each circuit of the cable shall exhibit an insulation resistance of 200 megohms or more when a potential of 500 ±50 volts DC is applied between each conductor or shield and all other conductors, shields, and connector shells of the cable, in accordance with MIL-STD-202, Method 302.

3.3.3 Engaging and disengaging forces.

3.3.3.1 Disengagement. An axial force of 100 pounds maximum, when applied between Extractor Weldment, BuWeps Dwg. 1660323, and Electrical

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Connector Receptacle Assembly LD 497942, shall disengage Cable Connector P2 (See 6.3.2.) from Electrical Connector Receptacle Assembly LD 497942 when fully engaged.

3.3.3.2 Non-disengagement. An axial force of 150 pounds minimum, when applied between Plug Housing Assembly BuWeps Dwg. 2169895 and Electrical Connector Receptacle Assembly LD 497942, shall not separate the connection between Cable Connector P2 and Electrical Connector Receptacle Assembly LD 497942 when fully engaged.

3.3.3.3 Engagement. An axial force of 80 pounds maximum shall be required to engage Cable Connector P2 with Electrical Connector Receptacle Assembly LD 497942 when Extractor Weldment BuWeps Dwg. 1660323 is in the retracted position.

3.4 Environmental requirements. Unless otherwise specified, the cable shall be capable of conforming to the performance and product characteristics of this specification after being exposed to the environments and all conditions specified herein.

3.4.1 Temperature.

3.4.1.1 Storage condition. The cable shall withstand temperatures from minus 65 to plus 160 degrees F. At conclusion of test, the connector inserts shall show no visual signs of cracking or deterioration and thereafter the cable shall meet the performance and product characteristics of this specification.

3.4.1.2 Operating condition. The cable shall meet the performance and product characteristics while at temperatures from 35 to 120 degrees F.

3.4.2 Dielectric voltage. The electrical connectors shall have a leakage current of not more than 1.0 milliampere when a potential of 1000 volts DC is applied between each conductor or shield and all other conductors, shields, and connector shells of the cable, in accordance with MIL-STD-202, Method 301.

3.4.3 Humidity. The cable shall withstand relative humidity up to 95 percent as specified in MIL-E-5272, Procedure I.

3.4.4 Salt spray. The cable shall withstand salt spray conditions as specified in MIL-E-5272, Procedure I.

3.5 Workmanship. The workmanship and finish shall be of sufficiently high grade to assure satisfactory operation consistent with the requirements of this specification. All parts used in the assembly shall be thoroughly clean and free of excess materials, chips, and loose or splattered foreign materials. The standards of workmanship exhibited in any approved preproduction sample,

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subject to any qualification stated in the Government' s notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection of the cable shall be classified as follows:

a. Preproduction and periodic production inspection, (See 4.5.)

b. Quality conformance inspection. (See 4.6.)

4.3 Sampling. (See 6.2.1.)

4.3.1 Lot size. The term "lot" shall mean "inspection lot", that is a collection of products submitted by a supplier for inspection. Unless otherwise specified in the contract or order, the number of products in each lot shall be three hundred (300).

4.3.2 Preproduction sample. A preproduction sample of ten (10) cables, manufactured in accordance with 3.2, shall be subjected to the preproduction tests detailed in 4.5. Acceptance of the preproduction sample shall be based on no defects in the sample. Further production of the cable by the supplier, prior to approval of the preproduction sample, shall be at the supplier' s risk.

4.3.3 Periodic production sample. From each succeeding lot of three hundred (300) or less, a random sample of five (5) cables shall be selected. These sample units shall be subjected to the tests of 4.5. The cables shall be considered satisfactory if they meet all specification and pertinent drawing requirements and withstand the tests specified in 4.5. (See Table I.) When a defective unit is found in any lot, acceptance of all units in the lot shall be withheld and the contractor shall:

a. Determine the cause of the defect.

b. Take the necessary action, with the concurrence of the procuring activity, to correct or eliminate the defect from all units in the lot and, if so directed by the procuring activity, from those units previously shipped.

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c. Submit to the procuring activity for approval, the proposed corrective action to prevent recurrence of the defect.

Table I. Preproduction and Periodic Production Test Sequence

Test	Paragraph Number	Test Quantities Preproduction	Periodic
Quality Conformance Inspection	4.6	10	5
Temperature, Storage	4.5.1.1	10	5
Dielectric Voltage	4.5.1.2	10	5
Quality Conformance Inspection	4.6	10	5
Humidity	4.5.1.3	10	-
Salt Spray	4.5.1.4	10	-
Quality Conformance Inspection	4.6	10	5
Temperature, Operating,	4.5.1.5		
Low Extreme	4.5.1.5.1	10	5
Quality Conformance Inspection	4.6	10	5
Temperature, Operating,	4.5.1.5		
High Extreme	4.5.1.5.2	10	5
Quality Conformance Inspection	4.6	10	5

4.3.4 Quality conformance inspection sampling. Samples shall be selected as specified in 4.6 and shall be used to determine the acceptability of the cable inspected as specified in 4.6.

4.4 Test equipment and test conditions.

4.4.1 Quality conformance inspection equipment. The following items of test equipment are required to perform the quality conformance inspection tests:

Equipment	Capability
a. Circuit Analyzer Test Set* Cable Assembly Mk 10 Mod 0	Refer to LD 615556. Operate in accordance with OD 24568.

*The low resistance measuring device (c) and the megohmmeter (d), with or without the Cable Tester (b), may be used in lieu of the Circuit Analyzer (a) for the electrical continuity test 4.6.1.1.1 and the insulation resistance test of 4.6.1.2.

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Equipment	Capability
b. Cable Tester, Cable Assembly Mk 10 Mod 0	Refer to LD 620138
c. Low resistance measuring device	0.00 to 0.5 ohm; accuracy plus or minus 5 percent
d. Megohmmeter	0 to 200 megohms; accuracy 10 percent at 200 megohms; 500 plus or minus 50 volts DC potential
e. Engagement and disengagement force measuring equipment	Apply and measure engaging and disengaging forces of 0 to 200 pounds; accuracy of plus or minus 5 percent

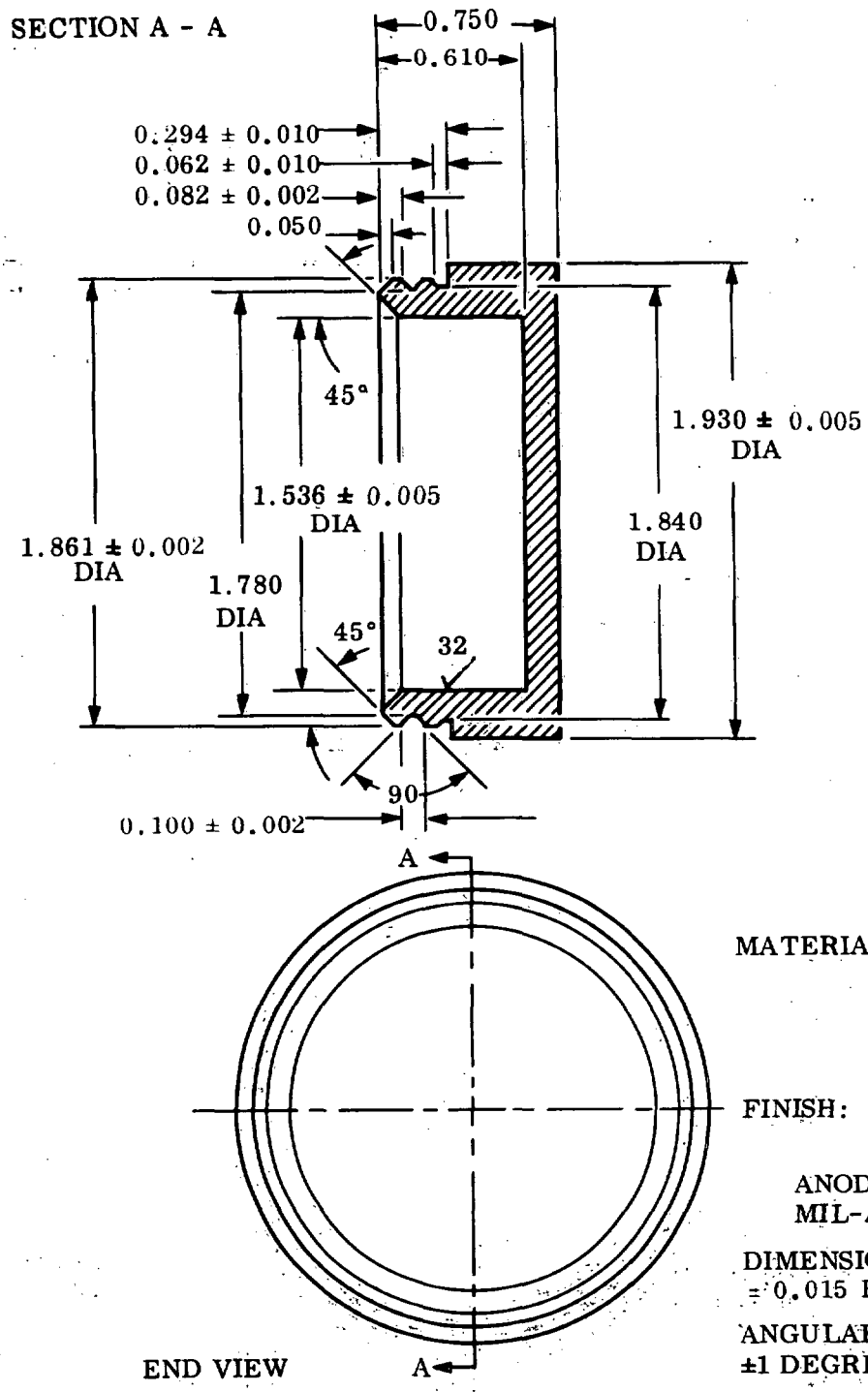
4.4.2 Preproduction and periodic production test equipment. The following items of test equipment, in addition to selected items of quality conformance inspection equipment and pieces of equipment recommended by applicable specifications or standards referenced within test procedure paragraphs, are required to perform the preproduction and periodic production tests:

Equipment	Capability
a. Temperature chamber	Maintain temperature at: Minus 65 plus 5 minus 0°F Plus 35 plus 5 minus 0° F Plus 120 plus 0 minus 5° F Plus 160 plus 0 minus 5°F
b. Electrical connector covers	Connector covers MS 25042-18 and MS 25042-28 with inside diameters of 0.932 to 0.954 and 1.525 to 1.547 respectively, and dummy connector cover, Figure 1.

4.4.3 Test conditions. Unless otherwise specified, the cable shall be subjected to the preproduction and periodic production tests and the quality conformance inspection tests under the following conditions:

a. Temperature	Room ambient (65 to 95 degrees F)
b. Altitude	Normal ground atmospheric pressure
c. Humidity (Relative)	Room ambient up to 95 percent

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MATERIAL: ALUMINUM ALLOY AA-2024-T4
ALTERNATE: QQ-A-225/6-T4

FINISH: 125 EXCEPT AS NOTED

ANODIZE PER MIL-A-8625 TYPE II

DIMENSIONAL TOLERANCE ± 0.015 EXCEPT AS NOTED

ANGULAR TOLERANCE ±1 DEGREE

Figure 1. Connector Cover

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4.5 Preproduction and periodic production inspections. The preproduction and periodic production samples shall be subjected to the tests specified in 4.5.1, after satisfactorily passing the quality conformance inspections of 4.6.

4.5.1 Test procedure. Subject the cable to the test sequence listed in Table I. Unless otherwise specified in the individual test paragraph, the quality conformance inspection as specified in Table I, shall be performed subsequent to environmental tests under test conditions specified in 4.4.3. Failure of the cable to pass any one of these tests shall constitute a defective unit.

4.5.1.1 Temperature, storage. The cable shall be subjected to a temperature of minus 65 plus 5 minus 0 degrees F for not less than 48 hours. The temperature shall be raised to plus 160 plus 0 minus 5 degrees F in not less than 2 hours and maintained at this temperature for not less than 48 hours. The unit shall then be subjected to room temperature for not less than 4 hours prior to performing subsequent tests.

4.5.1.2 Dielectric voltage. Subject the electrical connectors of the cable assembly to the dielectric withstanding voltage test in accordance with MIL-STD-202, Method 301. Apply a potential of 1000 volts DC for a duration of five seconds minimum between each conductor or shield and all other conductors, shields, and connector shells of the cable. The measured current during this test shall not exceed 1.0 milliampere. Thereafter the cable shall pass the quality conformance inspection tests of 4.6.

4.5.1.3 Humidity. Unless otherwise specified in the contract or order, this test shall be performed on preproduction sample units only. Install the electrical and dummy connector covers, specified in 4.4.2b, and subject the cable to Humidity Test, Procedure I, as specified in MIL-E-5272.

4.5.1.4 Salt spray. Unless otherwise specified in the contract or order, this test shall be performed on preproduction sample units only. Install the electrical and dummy connector covers, specified in 4.4.2b, on the cable, and subject the cable to the Salt Spray Test, Procedure I, as specified in MIL-E-5272 for a period of 50 hours.

4.5.1.5 Temperature, operating.

4.5.1.5.1 Low extreme. The cable shall be subjected to a temperature of 35 plus 5 minus 0 degrees F for not less than 2 hours and while at that temperature shall pass the quality conformance tests.

4.5.1.5.2 High extreme. The cable shall be subjected to a temperature of 120 plus 0 minus 5 degrees F for not less than 2 hours and while at this temperature shall pass the quality conformance tests.

4.6 Quality conformance inspection. Samples selected using the plans and procedures set forth in Table II of this specification, shall be subjected to the tests and examinations as described in 4.6.1.

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Table II. Quality Conformance Tests, Examinations and AQL

Test	Sampling Plan	AQL	Test Paragraph
Electrical continuity and circuit resistance	*	*	4.6.1.1
Insulation resistance	*	*	4.6.1.2
Engaging and disengaging force	**	**	4.6.1.3
Visual examination	*	*	4.6.1.4
Preservation, packaging, packing and marking	*	*	4.7

*One hundred percent inspection; defect is cause for rejection of individual unit.
**MIL-STD-105D Special Inspection Level S-3; zero defects accepted.

4.6.1 Test procedure. The cable shall be subjected to the tests and examinations as outlined herein. Unless otherwise specified, inability of the cable to pass the tests and examinations shall constitute a failure.

4.6.1.1 Electrical continuity and circuit resistance.

4.6.1.1.1 Electrical continuity. The electrical continuity of the cable circuits shall be checked using the automated circuit analyzer specified in 4.4.1a, calibrated to detect a resistance of 0.2 ohm. Circuits that are detected as having an indicated resistance exceeding 0.2 ohm shall be verified using the low resistance measuring device specified in 4.4.1c. The measurement shall be made directly across the pins of the cable circuit under question and shall not exceed 0.2 ohm.

4.6.1.1.2 Resistance of initiation circuits. The resistance of the initiation circuits listed below shall be measured using only the low resistance measuring device specified in 4.4.1c. The resistance shall not exceed 0.15 ohm.

Connectors J5 P2		Connectors J3 P2	
Pins H to	AA	Pins A to	BA
Pins J to	AC	Pins B to	BB
Pins R to	AB	Pins C to	BC

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4.6.1.2 Insulation resistance. Using the circuit analyzer equipment specified in 4.4.1a, measure the insulation resistance between each circuit of the cable in accordance with MIL-STD-202, Method 302, Condition B. When a potential of 500 plus or minus 50 volts DC is applied, the insulation resistance shall be 200 megohms.

4.6.1.3 Engaging and disengaging forces.

4.6.1.3.1 Disengagement. With Cable Connector P2 engaged with the Electrical Connector Receptacle Assembly LD 497942, an axial pulling force of 100 pounds maximum shall be applied between the Extractor Weldment BuWeps Dwg. 1660323, and the electrical connector receptacle assembly, using equipment specified in 4.4.1e, see Figure 2. Cable Connector P2 shall become disengaged from the electrical connector receptacle assembly.

4.6.1.3.2 Engagement. Using the equipment specified in 4.4.1e and with the extractor weldment in the retracted position, apply an axial force not to exceed 80 pounds between the Cable Connector P2 and the electrical connector receptacle assembly. The Cable Connector P2 shall become engaged with the electrical receptacle assembly.

4.6.1.3.3 Non-disengagement. Engage Cable Connector P2 with electrical connector receptacle assembly and place an adjustable hose clamp, one and three-fourths of an inch in diameter, minimum, around the plug housing assembly as shown in Figure 2 which will prevent the extractor weldment from releasing the connection. With the hose clamp in proper position, a force can be applied to the extractor weldment, which in turn is applied to the plug housing assembly via the hose clamp. Apply an axial force of 150 pounds between electrical connector receptacle assembly and the plug housing assembly, BuWeps Dwg. 2169895, via the extractor weldment using the equipment specified in 4.4.1e. The connector shall not separate under the applied load.

4.6.1.4 Visual examination. Visual examination shall be made on each cable submitted for delivery to the procuring agency. The cable shall be examined for conformance to the requirements of good workmanship in 3.5. Nonconformance shall be cause for rejection of individual unit.

4.7 Preservation, packaging, packing, and marking. The cable shall be examined for conformance to the prescribed preservation, packaging, packing, and marking requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be Level A or C as specified (See 6.2b and 6.4.)

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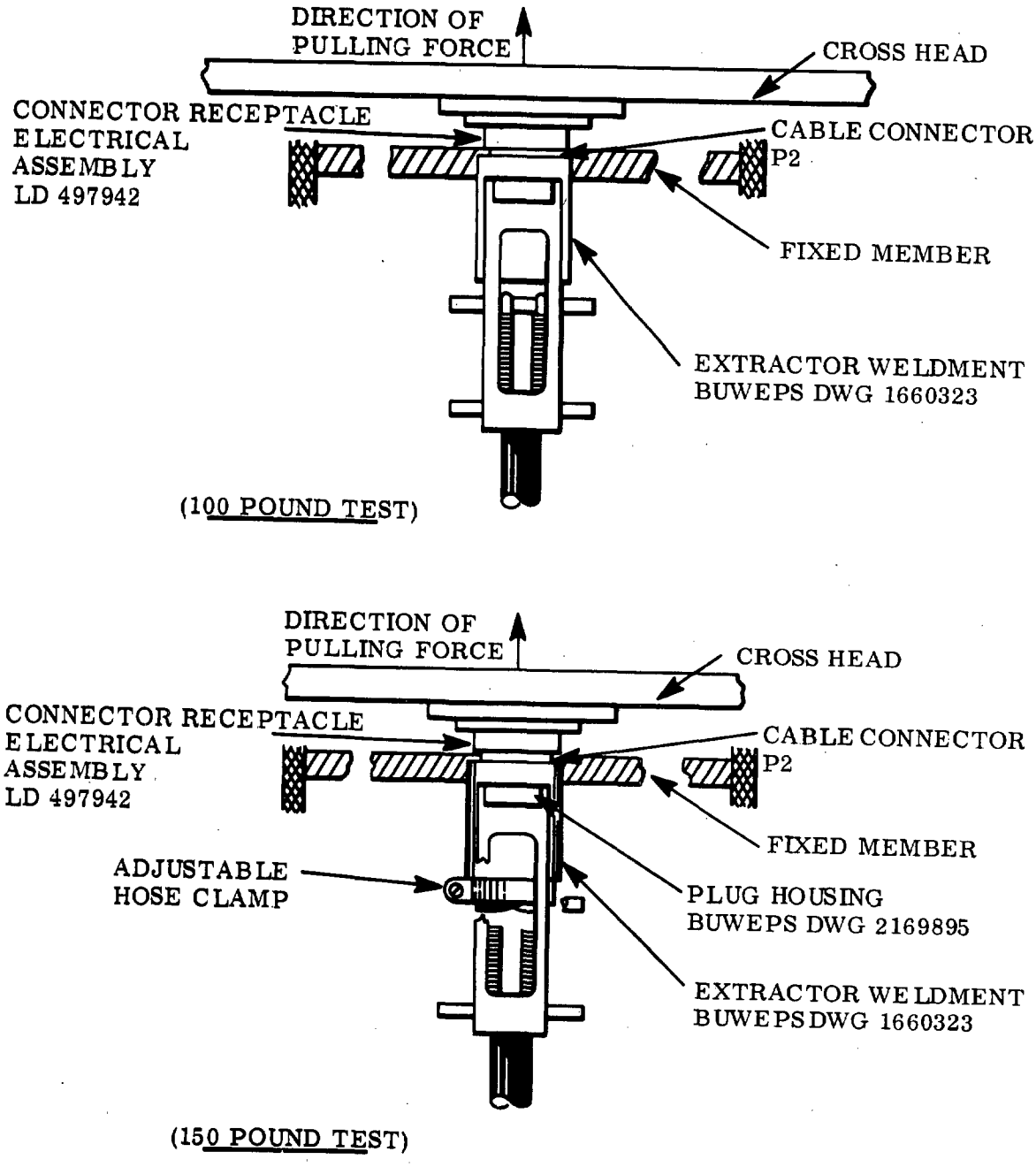


Figure 2. Disengaging Force Test

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5.1.1 Level A. Cleaning, drying, preservation, and packaging methods for Level A shipment shall be in accordance with BuWeps LD 540588.

5.1.2 Level C. Cleaning, drying, preservation, and packaging shall be in accordance with the supplier's commercial practice, when such practices have been approved by the procuring activity.

5.2 Packing. Packing shall be Level A or C as specified (See 6.2c and 6.4.)

5.2.1 Level A. Multiples of the cable preserved and packaged in accordance with 5.1.1 shall be packed in accordance with BuWeps LD 540589.

5.2.2 Level C. The cable, packaged as specified in paragraph 5.1.2, shall be packed to afford protection against damage during shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with common carrier regulations applicable to the mode of transportation.

5.3 Markings. In addition to the markings required by the contract or order, unit packages and shipping containers shall be marked in accordance with BuWeps LD 540589.

6. NOTES

6.1 Intended use and description.

6.1.1 Intended use. The cable is used to connect the ASROC missile to Launcher Mk 112.

6.1.2 Description. The cable is a multiconductor, electrical cable, which conducts electrical signals to the ASROC missile. The cable has a quick-disconnect device for disconnecting the cable from the missile at time of firing, allowing the cable to remain in the launcher.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Required level of preservation and packaging.

c. Required level of packing.

d. Any exceptions to this specification.

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6.2.1 Additional samples. Additional samples required because of failure of any sample(s) to pass the prescribed inspections shall be provided and inspected at the expense of the supplier.

6.3 Definitions. For the purpose of this specification, the following definitions have been established.

6.3.1 Operating. Operating is the condition of the cable when performing its required function.

6.3.2 Cable Connector P2. The Cable Connector P2 is defined as the portion of the cable which connects directly to the missile, and is the largest of the three connectors.

6.4 Preparation for delivery criteria. Criteria for use of proper level of preservation, packaging, and packing shall be as specified in FED-STD-102.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-C-82493(OS) CABLE ASSEMBLY MARK 10 MOD 0

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1426
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DEPARTMENT OF THE NAVY
Naval Ordnance Systems Command
Washington, D.C. 20360

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NAVAL ORDNANCE SYSTEMS COMMAND (ORD 9343)
DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20360

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